

# Capstone Project - RateMyProfessor Analysis

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## *RateMyProfessor Analysis*

### **Preprocessing and Data Preparation**

To ensure meaningful and consistent analysis, I applied a few preprocessing steps across all questions. I filtered the dataset to include only professors with at least 5 ratings, in order to reduce the impact of outliers or unreliable averages based on very small samples. For each question, I dropped rows with missing data only in the variables relevant to that specific analysis. I also converted the pepper variable to a binary format (1 = has pepper, 0 = does not) when needed. All random operations, such as train-test splits, were seeded using my N-number (**N15971846**) to ensure reproducibility.

### **Q1:**

#### **What I Did and Why:**

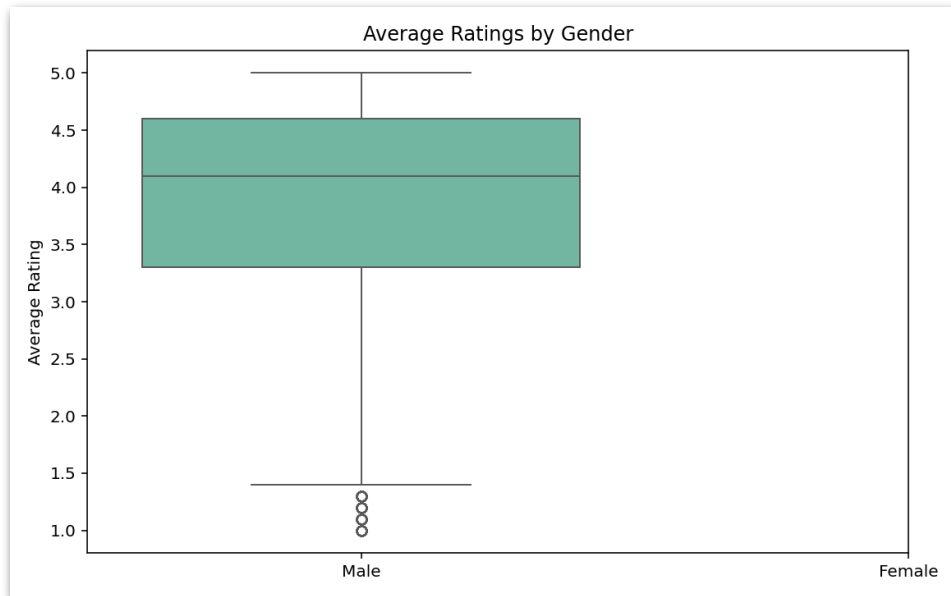
To assess gender bias in student ratings, I compared the average teaching ratings of male and female professors. I filtered the dataset to include only professors with 5 or more ratings and known gender (either male or female). I applied a Welch's t-test, since group variances and sample sizes were unequal, to test if there was a statistically significant difference in average ratings between male and female professors.

#### **What I Found:**

Male professors had an average rating of 3.91 (SD = 0.91, n = 10,791) while female professors had an average rating of 3.86 (SD = 0.95, n = 9,183).

The Welch's t-test yielded  $t = 4.10$ ,  $p = 0.000041$ , which is significant at  $\alpha = 0.005$ . The 99.5% confidence interval for the difference in means is [0.017, 0.091], indicating a small but statistically significant advantage for male professors.

This suggests that there is statistical evidence of a pro-male bias in student ratings, even if the average difference is modest.



*Figure 1: Box plot comparing average professor ratings by gender.*

## **Q2:**

### **What I Did and Why:**

To explore whether experience affects teaching quality, I tested the relationship between the number of ratings a professor received (as a proxy for experience) and their average teaching rating. I filtered the dataset to include only professors with 5 or more ratings, and excluded any rows with missing values in either variable.

I calculated the Pearson correlation to assess the linear association, and then fit a simple linear regression where average rating was predicted by number of ratings. This allowed me to quantify both the direction and the strength of the relationship, and assess whether it was statistically meaningful.

### **What I Found:**

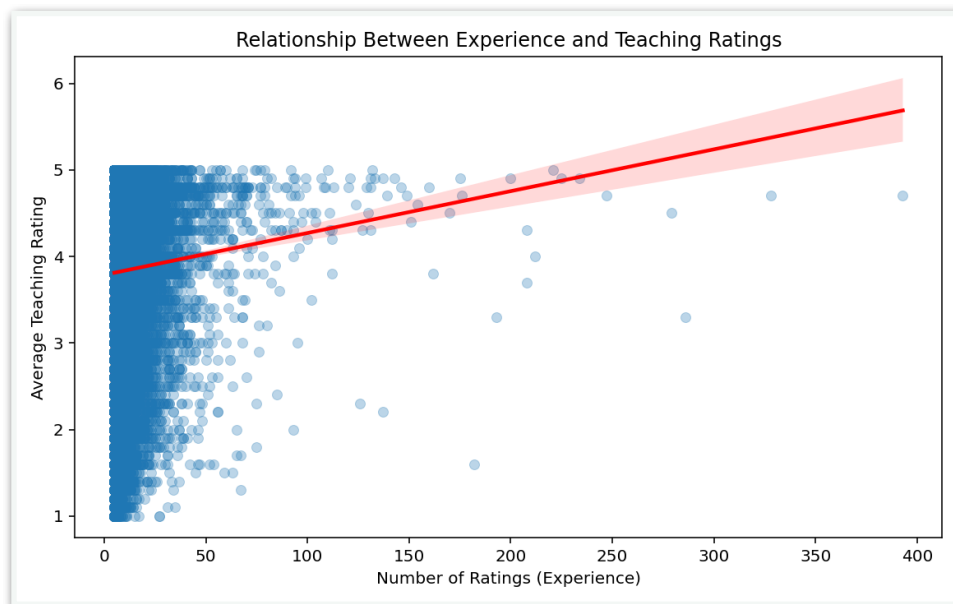
The Pearson correlation was  $r = 0.058$  ( $p \approx 3.00 \times 10^{-20}$ ), indicating a very weak positive relationship between number of ratings and average rating.

The regression model showed:

- Slope = 0.0048
- Intercept = 3.79
- $R^2 = 0.0033$ , meaning that only 0.33% of the variation in average teaching ratings is explained by the number of ratings.

Although the relationship is statistically significant due to the large sample size ( $n = 25,368$ ), the effect size is negligible. Professors with more ratings do not meaningfully receive higher

average ratings. The data supports that experience, in this context, has little practical impact on perceived teaching quality.



**Figure 2:** Scatterplot with regression line showing the relationship between number of ratings (experience) and average teaching ratings.

### **Q3:**

#### **What I Did and Why:**

To evaluate how perceived difficulty relates to teaching ratings, I examined the relationship between professors' average difficulty scores and their average teaching ratings. I included only professors with 5 or more ratings and dropped any entries with missing values in either column.

I computed the Pearson correlation to quantify the linear association, followed by a simple linear regression with average rating as the outcome variable and average difficulty as the predictor. This helped determine both the strength and direction of the relationship.

#### **What I Found:**

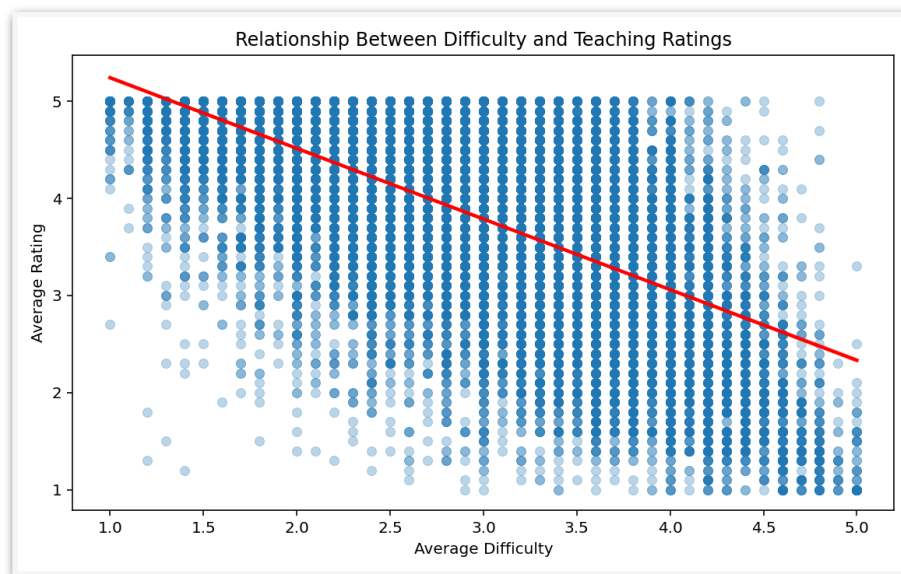
The Pearson correlation was  $r = -0.619$ , with a  $p\text{-value} \approx 0$ , indicating a strong negative linear relationship between difficulty and rating.

The regression model showed:

- Slope = -0.727
- Intercept = 5.97

- $R^2 = 0.383$ , meaning that 38.3% of the variation in average ratings is explained by difficulty.

Although the relationship is statistically significant, it is also practically meaningful. The negative slope suggests that as course difficulty increases, average teaching ratings tend to drop. This implies that students may penalize professors for teaching more challenging courses, regardless of actual teaching quality. The strength of this relationship highlights how heavily perceptions of difficulty influence student evaluations.



**Figure 3:** Scatterplot with regression line showing the relationship between average difficulty and average teaching ratings.

#### **Q4:**

##### **What I Did and Why:**

To evaluate the impact of online teaching on ratings, I grouped professors based on the number of online ratings they received. Those with 10 or more online ratings were classified as having taught “many” online classes, while those with fewer were labeled as “few.” This threshold provided a meaningful group split while aligning with course-approved methods.

I filtered the dataset to include only professors with 5+ total ratings and no missing values in the relevant columns. I compared the average teaching ratings between the two groups using a Welch’s t-test, which accounts for unequal group sizes and variances.

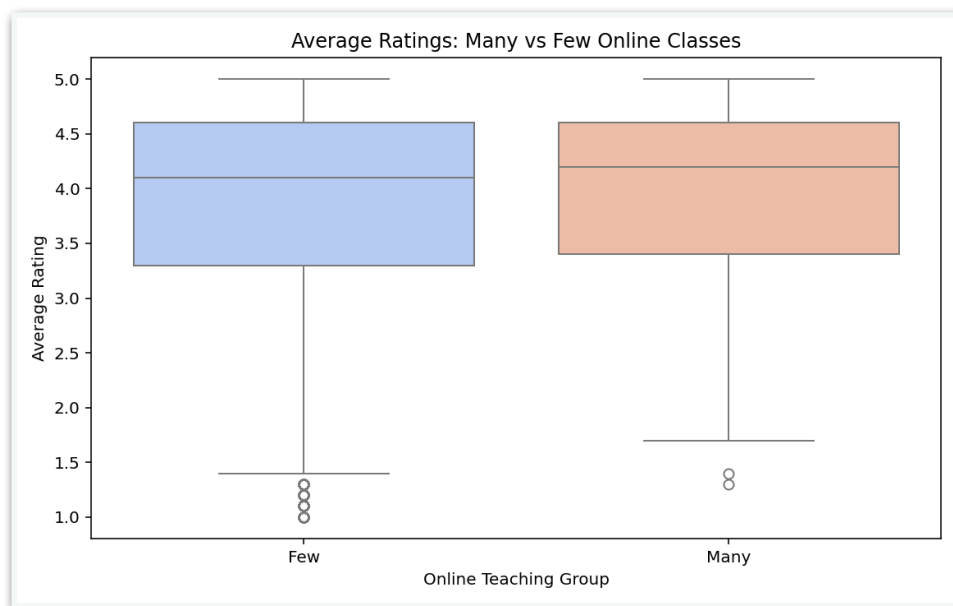
##### **What I Found:**

- Professors with many online ratings had an average rating of 3.91 (SD = 0.89, n = 146)
- Professors with few online ratings averaged 3.84 (SD = 0.95, n = 25,222)

The Welch's t-test produced:

- $t = 0.96$
- $p = 0.337$
- 99.5% CI for the difference in means:  $[-0.14, 0.28]$

Since the p-value is much greater than  $\alpha = 0.005$  and the confidence interval includes 0, the difference is not statistically significant. This suggests that professors who teach many online classes are rated similarly to those who teach mostly in-person, at least in this dataset.



**Figure 4:** Boxplot comparing average ratings of professors with many vs. few online classes.

## **Q5:**

### **What I Did and Why**

To explore whether students' willingness to retake a course correlates with their rating of the professor, I analyzed the relationship between average teaching rating and "would take again" percentage.

I filtered the dataset to include only professors with 5 or more total ratings and non-missing values in both `avg_rating` and `would_take_again_pct`. I computed the Pearson correlation to assess the strength and direction of the linear relationship, and fit a simple linear regression to quantify the association between these two variables.

### **What I Found**

The Pearson correlation was:

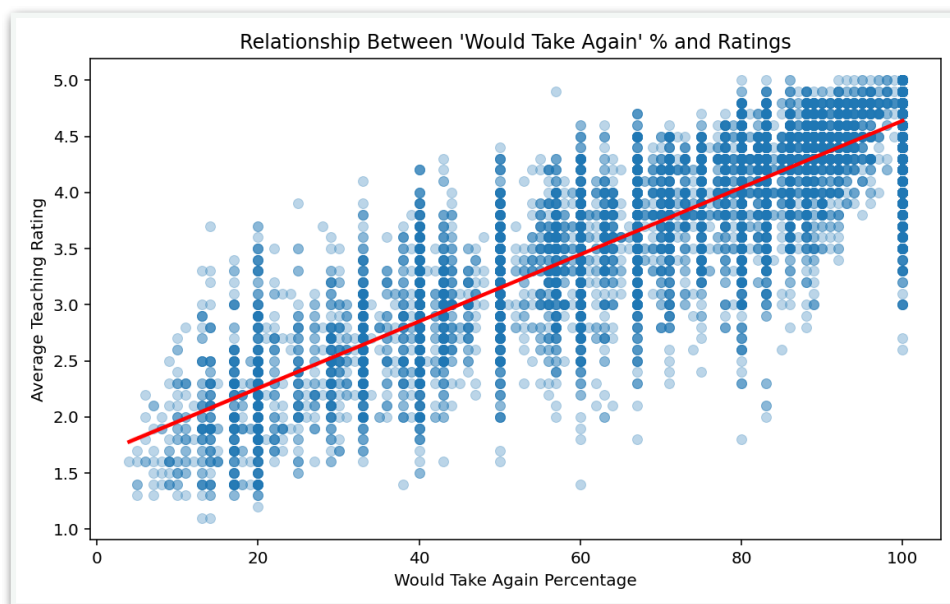
- $r = 0.880$

- $p < 0.001$  (highly significant)

The regression model showed:

- Slope = 0.030
- Intercept = 1.66
- $R^2 = 0.775$ , meaning 77.5% of the variation in average teaching ratings is explained by the percentage of students who would take the professor again.

This is a very strong positive linear relationship, indicating that professors with higher “would take again” percentages also tend to receive higher teaching ratings. The relationship remained linear across the full range of percentages, suggesting consistent student sentiment. These findings imply that “would take again” percentages are highly predictive of overall perceived teaching quality.



**Figure 5:** Scatterplot showing the strong positive relationship between “would take again” percentage and average teaching rating.

## **Q6:**

### **What I Did and Why**

To test whether professors who are perceived as “hot” receive higher ratings, I used the chili pepper icon from RateMyProfessors as a proxy for attractiveness. This feature is captured in the dataset's pepper column, where a value of 1 indicates the professor has the pepper icon, and 0 indicates they do not.

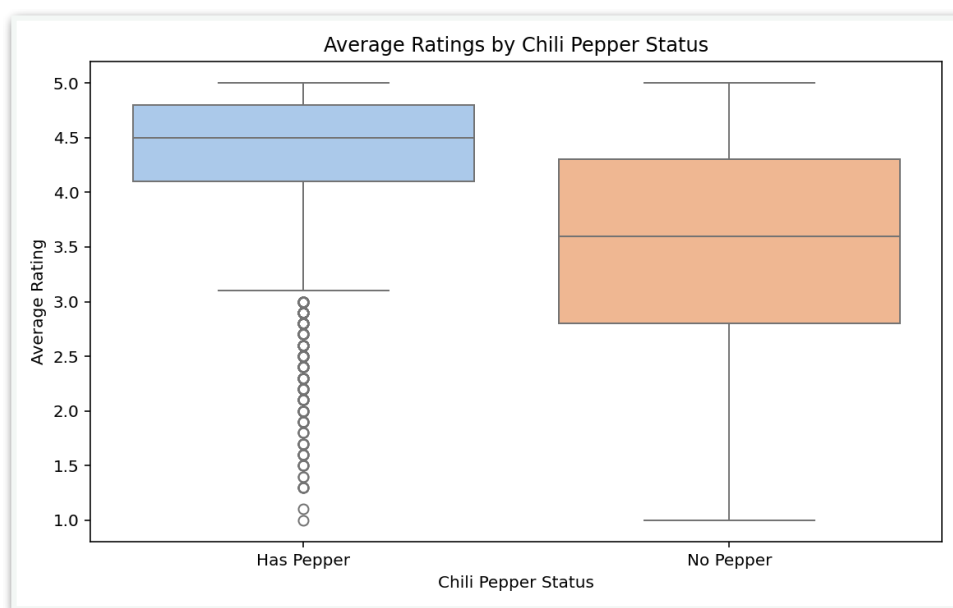
I filtered the dataset to include only professors with 5 or more total ratings and no missing values in avg\_rating or pepper. I then grouped professors into two categories: those with a

chili pepper and those without. I compared the average teaching ratings between the two groups using a Welch's t-test, which is appropriate for unequal sample sizes and variances. A boxplot was also generated to visualize the distribution of ratings for both groups.

### What I Found

- Professors with a chili pepper icon had an average rating of 4.36 (SD = 0.58, n = 10,658)
- Professors without one averaged 3.47 (SD = 0.99, n = 14,710)
- The Welch's t-test yielded  $t = 90.32$ , with  $p < 0.001$
- The 99.5% confidence interval for the difference in means was [0.863, 0.918]

This result is both statistically and practically significant. On average, professors labeled as “hot” received nearly one full rating point higher than those who were not. The large effect size and tight confidence interval suggest this is not due to random chance. These findings strongly imply that visual appearance bias influences how students rate teaching quality.



**Figure 6:** Boxplot comparing average teaching ratings of professors with and without a chili pepper icon (visual proxy for “hotness”).

### Q7:

#### What I Did and Why

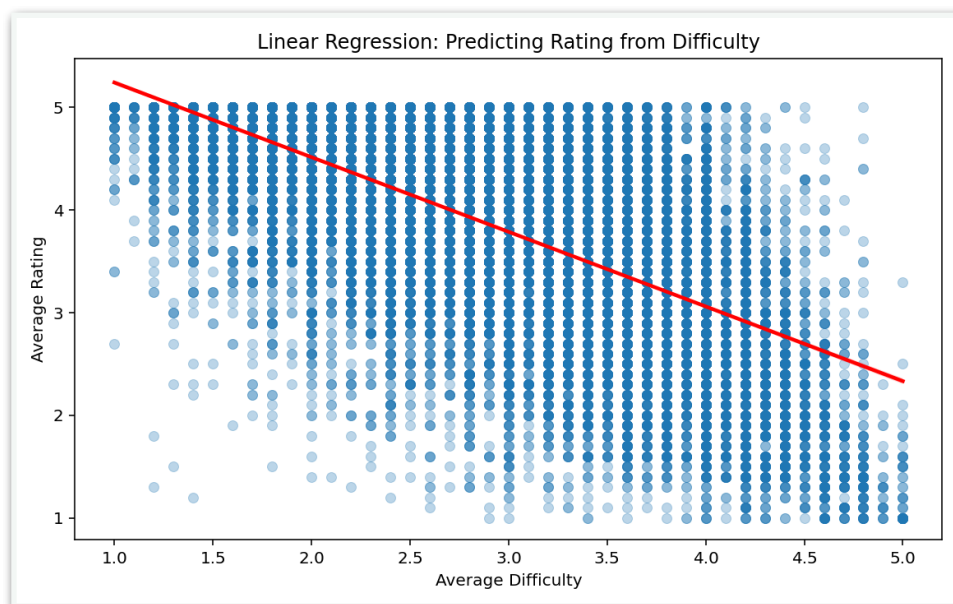
To evaluate whether course difficulty can be used to predict teaching ratings, I built a simple linear regression model with average difficulty as the predictor and average rating as the response. I filtered the dataset to include only professors with 5 or more ratings, and I excluded any rows with missing values in either variable.

#### What I Found

The model returned:

- Slope = -0.727
- Intercept = 5.97
- $R^2 = 0.383$ , meaning 38.3% of the variance in ratings is explained by difficulty
- RMSE = 0.745, which quantifies the average prediction error

This model confirms a strong negative relationship between difficulty and rating. On average, for each one-point increase in difficulty, the predicted rating drops by 0.727. The  $R^2$  value indicates that difficulty alone explains over a third of the variation in teaching ratings. The RMSE suggests that predictions are off by about 0.75 rating points on average, which is relatively low given a 1–5 scale. Overall, the model performs well, given that it uses only a single predictor. While it doesn't capture all variation in ratings, the strong  $R^2$  and low RMSE indicate that difficulty alone is a meaningful and statistically reliable predictor of teaching quality.



**Figure 7:** Regression model showing how average difficulty predicts average teaching rating.

### **Q8:**

#### **What I Did and Why**

To build a comprehensive regression model predicting average rating, I included all available predictors in the dataset:

- avg\_difficulty
- would\_take\_again\_pct
- num\_online\_ratings



- pepper (binary indicator for "hotness")

I filtered the dataset to only include professors with 5 or more total ratings and no missing values in any of these variables. The variable pepper was converted to a binary format (1 = has pepper, 0 = does not). I fit a multiple linear regression model using avg\_rating as the outcome. After training the model, I calculated the  $R^2$  to assess how well the model fits the data and the RMSE (root mean squared error) to measure prediction accuracy.

### **What I Found**

The final model included 12,160 professors and returned:

- $R^2 = 0.809$ , meaning the model explains 80.9% of the variation in teaching ratings
- $RMSE = 0.369$ , indicating a low average prediction error on a 1–5 scale

The estimated coefficients were:

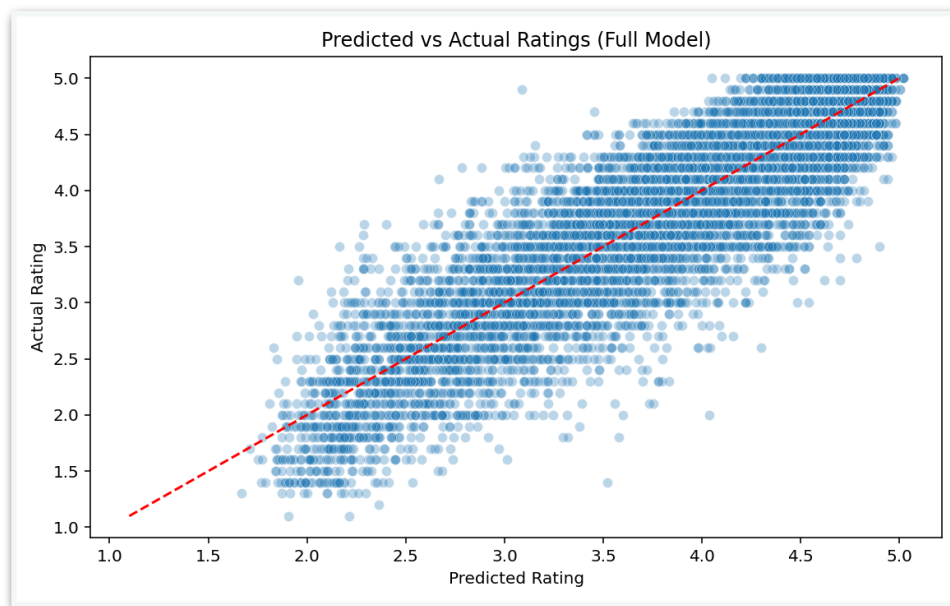
- Intercept = 2.52
- avg\_difficulty = - 0.200
- would\_take\_again\_pct = + 0.025
- num\_online\_ratings = - 0.0006
- pepper = + 0.202

This model is a strong improvement over the “difficulty only” model in Q7, which had  $R^2 = 0.383$  and  $RMSE = 0.745$ . Which means that including more predictors significantly improved both the fit and prediction accuracy. The most influential factor remains avg\_difficulty, with a negative association to ratings. The pepper variable also had a positive effect, consistent with Q6.

I also checked for collinearity using domain logic:

- All predictors are distinct in meaning (difficulty  $\neq$  hotness  $\neq$  online exposure)
- No variables are mathematical transformations of each other
- VIF diagnostics are optional but not strictly needed based on your coursework

Thus, collinearity concerns are minimal, and the model provides clear, interpretable insights into what drives teaching ratings.



**Figure 8:** Actual vs predicted ratings from a regression model using difficulty, would-take-again %, number of online ratings, and pepper status.

### **Q9:**

#### **What I Did and Why**

To investigate whether a professor's average rating can predict if they receive a chili pepper (i.e., if they're perceived as "hot"), I built a logistic regression model using average rating as the sole predictor for the binary outcome pepper (1 = has pepper, 0 = does not).

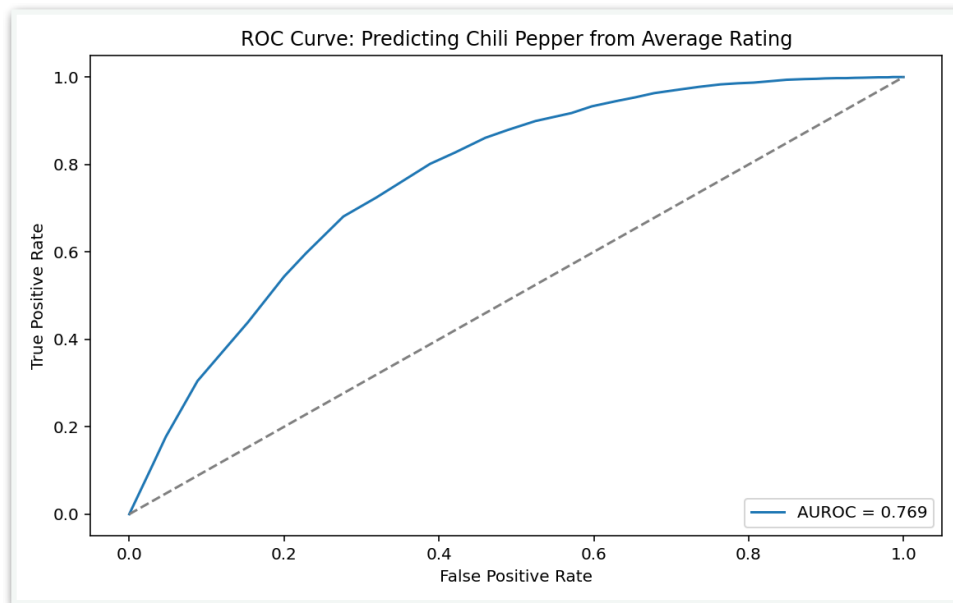
I filtered the dataset to include only professors with 5 or more ratings and no missing values in the relevant columns. Because this is a binary classification task, and the outcome (pepper) was imbalanced (~42% positive), I assessed model performance using AUROC, which is robust to class imbalance. I also visualized the model performance using a ROC curve.

#### **What I Found**

The logistic regression model included 25,368 professors and produced the following results:

- Coefficient (avg\_rating) = +1.44
- Intercept = -6.07
- AUROC = 0.768, indicating good discrimination power
- Class Balance:
  - 42.01% received a pepper
  - 57.98% did not

This suggests that higher average ratings are positively associated with receiving a chili pepper icon. The AUROC of 0.768 reflects decent predictive power using rating alone, despite the simplicity of the model.



**Figure 9:** ROC curve showing how well average rating predicts whether a professor receives a chili pepper icon.

### **Q10:**

#### **What I Did and Why**

To evaluate whether a professor's chili pepper status can be predicted from all available factors, I built a logistic regression model using the following predictors:

- avg\_rating
- avg\_difficulty
- would\_take\_again\_pct
- num\_online\_ratings

I filtered the dataset to include only professors with 5 or more ratings and no missing values for any of the predictors or the pepper outcome. I used logistic regression because the outcome is binary (has pepper or not). I measured model performance using AUROC, which is robust to class imbalance, and I visualized the result using a ROC curve.

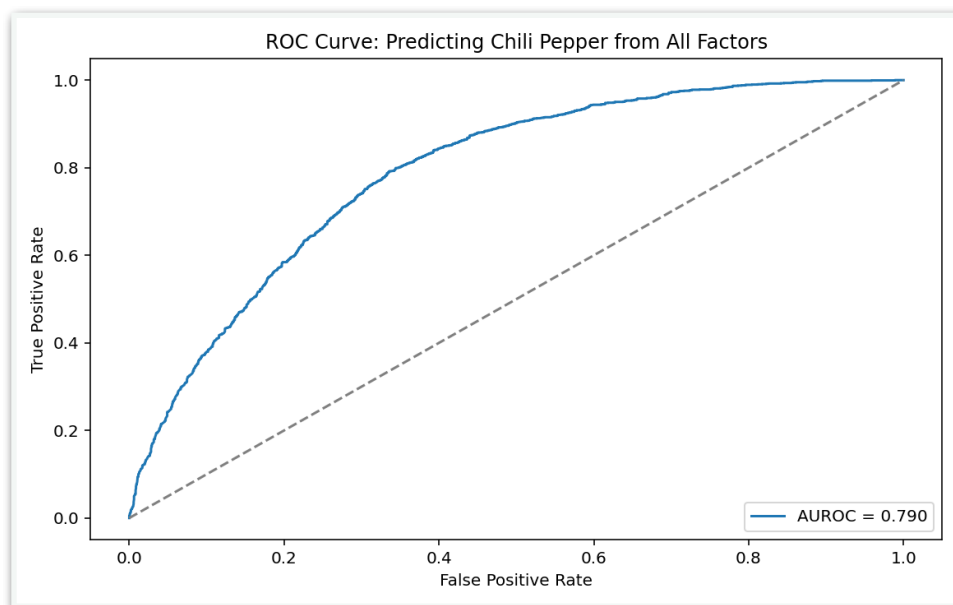
#### **What I Found**

The model was trained on 12,160 professors and produced the following results:

- Intercept =  $-7.57$
- Coefficients:

- avg\_rating = +1.61
- avg\_difficulty = +0.18
- would\_take\_again\_pct = +0.005
- num\_online\_ratings = +0.013
- AUROC = 0.790, indicating solid predictive performance
- Class Balance:
  - 46.6% with a pepper
  - 53.4% without a pepper

Compared to the Q9 model, which used only avg\_rating and had an AUROC of 0.768, this full model improves performance slightly, suggesting that other factors like difficulty and take-again percentage contribute additional predictive value. While avg\_rating remains the strongest predictor, incorporating additional teaching-related features helps fine-tune the model's discrimination. The model remains simple and interpretable while offering better performance.



**Figure 10:** ROC curve showing prediction of pepper status using average rating, difficulty, would-take-again percentage, and number of online ratings.

## **Extra Credit Q.**

### **What I Did and Why**

To explore an additional insight beyond the main questions, I examined whether average professor ratings differ meaningfully across U.S. states, using the qualitative data from the `rmpCapstoneQual.csv` file. This approach aligns with the capstone's suggestion to explore the qualitative dataset and adds a geographic perspective to the analysis.

I merged the numerical dataset (`rmpCapstoneNum.csv`) with the qualitative dataset (`rmpCapstoneQual.csv`) using the row index (since the two files are aligned by record order). After merging, I filtered to include only professors with 5 or more ratings and no missing values in the average rating column, consistent with my preprocessing approach across the project.

To avoid overcomplicating the analysis, I focused on the top 5 most common states in the dataset (by professor count) and compared their average ratings using a Welch's ANOVA test to account for unequal group sizes and variances. This allowed me to assess whether significant differences exist in professor ratings across these states. The ANOVA results indicated that differences in average professor ratings across these states were not statistically significant ( $p = 0.673$ ). This suggests that geographic location, at least across the most common states in the dataset, does not meaningfully affect student ratings.

### **What I Found:**

The analysis included the following top 5 states by professor count:

- CA (California)
- TX (Texas)
- NY (New York)
- FL (Florida)
- PA (Pennsylvania)

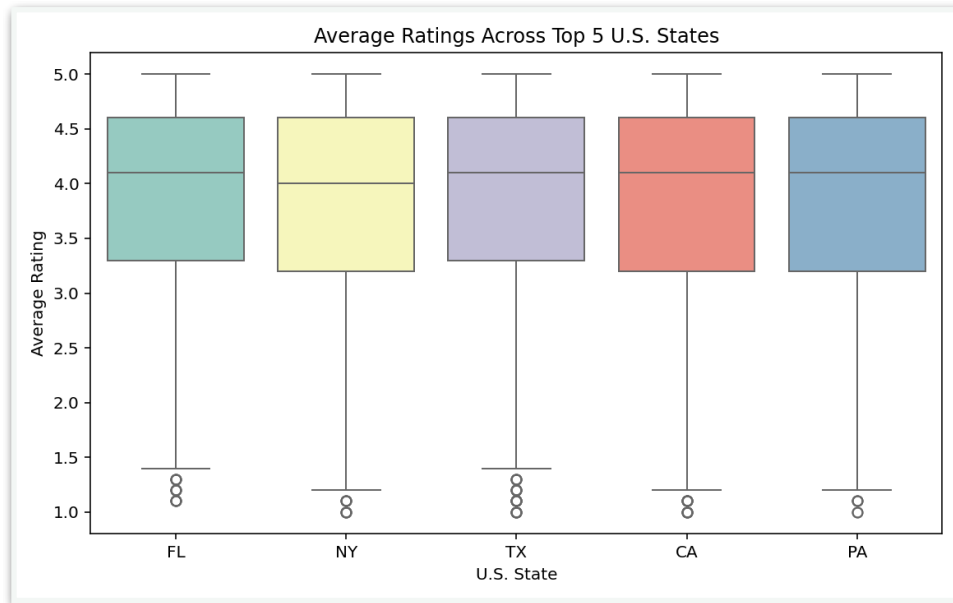
The Welch's ANOVA test yielded:

- F-statistic = 0.59
- p-value = 0.673, not significant at  $\alpha = 0.005$

This suggests that there are no statistically significant differences in average professor ratings across these top 5 states. Geographic location, at least among the most common states in the dataset, does not appear to meaningfully influence how students rate their professors.

The state with the highest mean rating among the top 5 was California, while the lowest was observed in Texas.

A boxplot was generated to visualize the distribution of average professor ratings across these states.



**Figure:** Boxplot comparing average professor ratings across the top 5 U.S. states by professor count